PSYC4411

BUILD A DEFENSIBLE CLASSIFIER

WEEK 6 CHALLENGE LAB

Trees, Ensembles & Real Mental Health Data

TODAY'S CHALLENGE

- **Build classifiers** for elevated depression (PHQ-9 ≥ 5)
  - Baseline → Logistic Regression → Decision Tree → Random Forest
- **Report proper metrics** — not just accuracy
  - Precision, recall, F1, AUC, confusion matrix
- **Justify your threshold** — why 0.5?
  - Try at least one alternative and compare
- **Present your results** (1 slide, ~3 min)
  - Best model + metrics, threshold justification, feature importance, ethical consideration, refactoring win

THE DATASET: BC COVID SLEEP & WELL-BEING

- **3 CSV files** to merge:
  - `daily_survey.csv` — 52K daily entries
  - `demographics.csv` — age, sex
  - `round1_assessment.csv` — Big Five, GAD-7
- **After merging:** ~836 participants
- **Target:** PHQ-9 ≥ 5 (elevated depression)
  - ~54.5% elevated / ~45.5% minimal — nearly balanced
- **21 features:** sleep, mood, personality, demographics, anxiety

Data prep is done for you! The starter notebook loads, aggregates, and merges the data in cells 1–8. You start modelling from cell 9.

Key features: PANAS negative affect, GAD-7 anxiety, stress coping, sadness, isolation, sleep latency, personality (Big Five)

NEW LLM SKILL: REFACTORING

Week 2: **Prompting** · Week 4: **Debugging** · Week 6: **Refactoring**

Weak

"Clean up my code."

Strong

"Refactor to: (1) separate loading from modelling, (2) add shape assertions after merges, (3) create a reusable `evaluate_model()` function, (4) add docstrings. Keep logic identical."

Refactoring = making code cleaner without changing what it does. Aim for code someone else could read and understand.

GETTING STARTED

STEPS

1. Open `starter.ipynb` or `starter.py`
2. Run cells 1–8 (data is loaded & merged)
3. Ask your AI to **plan first**
4. Build: Baseline → LogReg → Tree → Forest
5. Evaluate, analyse, refactor, present

TARGET NUMBERS

Baseline accuracy: ~54.5%

Good model accuracy: ~80%

Good model AUC: ~0.90

Top predictors: PANAS negative affect, GAD-7, stress coping